

Curriculum Vitae/Resume - Derek Yang

SUGS Computer Science & Engineering M.S.E.

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Education

University of Michigan

Aug 2022 - Dec 2025

Bachelor of Science in Engineering in Computer Science, Minor in Business, UX Design

Ann Arbor, MI

- **GPA:** 3.97
- **Teaching Assistant:** Software Engineering
- **Awards:** AR Winner (XRMidwest Hackathon), Best Project (XR Exhibition), CLAWS Hackathon Winner, Best Environment Award (XR at Michigan Summit 2023)

Experience

VR Software Engineering Intern

Rockville, MD

BAE Systems, Inc.

May 2025 - Aug 2025

- Built a **VR** visualization tool for the Ohio-class submarine that transforms **MATLAB-generated data** into immersive **3D simulations** through **JSON-based data pipelines** developed in **Lua**, enabling real-time comprehension of complex system behaviors
- Researched methods for integrating **LLM-based context awareness** into an acronym glossary application, evaluating approaches for **secure data handling** and improved contextual definition accuracy

Software Engineer Intern

Remote

Next Play

May 2024 - Aug 2024

- Developed **React** components in **TypeScript** for a fantasy sports app, enabling users to edit lineups by dynamically updating selections and fetching MLB player data from a **PostgreSQL** database, enhancing real-time interactivity
- Performed **quality assurance** using **Postman** to **mock test APIs** and validate **PostgreSQL** database interactions with user data, **ensuring data integrity** and **reducing potential backend errors** before deployment
- Tested components with **Docker** to ensure reliability and scalability, contributing to an app launch with 500+ users

AR Research Assistant

Ann Arbor, MI

University of Michigan Robotics Department

May 2023 - Aug 2023

- Researched shared mental models in human-robot interactions by developing an **AR** color-sorting task on Magic Leap 2 using **C#** and **Unity SDK**'s eye and hand tracking **APIs**, demonstrating mixed-agent team collaboration
- Integrated **ROS2** in **Unity** to automate **AR** robots, implementing a **Pub/Sub** system to simulate shared mental models and compare the effects of collaborative versus independent agent behaviors on task coordination

AI Lead Developer / AR Integration Coordinator (Executive Board)

Ann Arbor, MI

NASA, CLAWS-UM

Apr 2023 - Present

- Constructed a **full-stack AR** interface using **Unity** and **Microsoft MRTK** for **NASA**'s national competition, presenting a more efficient astronaut EVA solution at Johnson Space Center
- Implemented geosampling and astronaut communication by developing the **frontend** in **Unity** and **backend** in **C#** with **APIs**, **Pub/Sub** design, and **state machine**, enabling astronauts to perform mission-critical tasks

Projects

VS Code Contribution | TypeScript, Git, TravisCI, Azure DevOps

Nov 2024 - Dec 2024

- Resolved **CSS** error handling in embedded **HTML** documents by enhancing **HTML** language server with **TypeScript**, following the repository's **coding standards** to ensure quality and consistency
- Diagnosed and fixed faults in **1.5M+** line codebase, ensuring quality through **integration tests** in Azure DevOps
- Collaborated with **2,000+** contributor open-source community, refining PRs through **code review** with maintainers

Arcane Spell Language - XR Exhibition Winner | C#, Unity, VR, Git

Nov 2024 - Dec 2024

- Implemented **VR** ASL learning app by integrating **hand and gesture tracking** with the **Meta Interaction SDK** and **multiplayer** with **Unity Netcode**, allowing users to collaboratively learn the ASL alphabet
- Conducted **4 user tests** and triaged issues in **Jira**, driving **agile iterative sprints** to improve user experience

Technical Skills

Languages: C/C++, Python, JavaScript, TypeScript, SQL, C#, Java, HTML/CSS, R, Assembly

Frameworks: React, Node.js, Flask, MySQL, SQLite, PostgreSQL

Tools: Git, GitHub, Docker, AWS, Linux, TravisCI, Azure DevOps, VS Code, Visual Studio, Unity, Unreal Engine